VBA Final Exam

Complete both problems in desktop Excel using the attached workbook. Detect the last data row at runtime, qualify worksheet references, and preserve the source sheets. A second run must replace the previous output. Submit a ZIP named with your full name containing the .xlsm workbook and exported VBA modules.

1. Calculate restock quantities (50 points)

Use Stock. Write ReorderQty(onHand As Long, target As Long) As Long to return the quantity needed to reach the target, with a minimum of zero. Write a public Sub named BuildRestockList. Trim and uppercase each SKU. Reject and count rows with a blank SKU or blank, nonnumeric, fractional, or negative quantities. Assume valid quantities fit in a Long. Call the function for each valid row. Write items with a positive reorder quantity to Restock, including SKU, OnHand, Target, and ReorderQty. Sort entire records by reorder quantity from highest to lowest. Display total units to order and the rejected row count.

2. Report open balances (50 points)

Use Invoices and the region in Controls!B2. Amounts are valid nonnegative numbers. Write OpenBalance(invoiced As Double, paid As Double) As Double to return the unpaid amount, with a minimum of zero. Write a public Sub named BuildBalanceReport. Match the selected region after trimming spaces and ignoring case. Call the function for each matching row. Write only invoices with a positive balance to Balances. Include invoice ID, region, invoiced amount, paid amount, and open balance. Use FormulaR1C1 to calculate each invoice's share of the total open balance. Display the count and total balance. If no rows qualify, show zero for both totals and leave the detail area empty.

VBA-final-exam.xlsx